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The coupled mixing action of the jet mixer and swirl mixer: An novel static micromixer

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ABSTRACT

The development of high-performance static mixers is a critical research topic in process intensification in the process industry. This research combines computational fluid dynamics simulation and experimental testing to design a new type of static micromixer: First, a variable-diameter swirl shear mixer was designed and its structure was optimized. Measured by Malvern laser particle size analyzer, the median particle size decreased from 41 μm to 29 μm in the designed micromixer compared to that of traditional static mixers. Subsequently, the dispersed phase injection method was upgraded to a jet injection method, and the static micromixer was designed with a combination of an ejector and a variable-diameter swirl shear mixer. The median particle size of the dispersed droplets further decreased to 23 μm, achieving a dispersion effect at 2100−2400 rpm of the dynamic high-shear mixer. The micromixer greatly improved the static mixing effect and may replace dynamic high-shear mixers in the future.

1. Introduction

Liquid–liquid heterogeneous mixing and dispersion is a process wherein two incompatible or completely insoluble liquids are uniformly mixed (Thakur et al., 2003). This method is widely used in polymer blending, chemical reactions, food processing, biopharmaceuticals, cosmetics, and other fields. Especially in the chemical industry, the mixer can be used for droplet breaking and mixing, increasing the contact interface, for solvent extraction, washing, and other mass transfer processes; and intensified mixing, for polymerization, alkylation, and other liquid–liquid reactions. Generally, the better the liquid–liquid mixing effect, the better the performance of the above process. The mixer is divided into static mixers and dynamic mixers. Static mixers such as Venturi jet mixing (Liu et al., 2021), SK static mixer, SV static mixer (Montante et al., 2016), impinging stream mixers, etc., have simple structure and low pressure drop, but the mixing effect is not good. The dispersed median diameter of droplets is usually greater than 40 μm, resulting in low extraction or reaction process performance. Dynamic mixers such as high-shear mixers such as kettle agitators (Zhao et al., 2018) and emulsification pumps (Akiti et al., 2020; Bayareh et al., 2020; Chabanon et al., 2017; Cortada-Garcia et al., 2018; Gyurik et al., 2020; Hert and Rodgers, 2017; Jegatheeswaran et al., 2018; Kundra et al., 2020; Lebaz and Sheibat-Othman, 2019; Liu et al., 2021; Montante et al., 2016; Sabaghian et al., 2018; Scala et al., 2020; Sun et al., 2020; Thakur et al., 2003; Vikhansky, 2020; We et al., 2021; Zhao et al., 2018; Zhuang et al., 2020; Sutherand, 1784), such mixers are characterized by high local energy dissipation caused by the meshing of the rotor and stator (Hert and Rodgers, 2017). The mixing effect is perfect, and the median particle size can be less than 20 μm. However, it is not suitable for applications involving high pressure or continuous production processes (Cortada-Garcia et al., 2018). At present, there is still a lack of static mixers with mixing performance comparable to dynamic high-shear mixers. We have found that coupling multiple types of static mixers is an effective means to improve the mixing performance. In particular, the dispersed phase spray injection coupled with the swirling shear mixing method, can construct a static mixer with micro-mixing performance.

Although static mixers were not generally established in the process industries until the 1970s, the first patent application can be traced back to 1874 (Sutherand, 1784). Sutherland applied for a patent to mix air with fuel gas through multi-layer static equipment. In 1970, Kenneth Company of the United States obtained a license for invention from Arthur D. Little Company and achieved the industrial production application of a static mixer for the first time worldwide. In 1973, the Sulzer

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static mixer was invented by a Swiss company; this became a milestone in the research and application of static mixers (Jegatheeswaran et al., 2018; Chabanon et al., 2017). In the 21st century, with the development of computational fluid dynamics (CFD), increasing number of researchers attempted to optimize the design of static mixers using the internal flow field of static mixers. Sun et al. (2020) designed a Chinese-knot type static mixer and studied the flow and heat transfer characteristics under high viscosity using CFD simulation. Zhuang et al. (2020) studied a new type of double-vortex static mixer for gas mixing using CFD and verified the simulation results using particle image velocimetry (PIV). Vikhansky (2020) studied the emulsification process in a highly efficient vortex mixer under turbulent conditions using the adaptive multi-scale method. With the continuous development of CFD technology, the internal flow field simulation of static mixers is constantly developing (Gyurik et al., 2020). Current research on static mixers is predominantly focused on structural optimization and application expansion (Kundra et al., 2020; Bayareh et al., 2020; We et al., 2021; Scala et al., 2020; Sabaghian et al., 2018; Lebaz and Sheibat-Othman, 2019), but there remains room for improvement regarding the structural parameters and injection mode of the static mixer. This constitutes the scope of our further research.

In this study, a variable-diameter swirl shear mixer was designed based on a heterogeneous system of diesel/water. The particle size distribution and turbidity sedimentation curve of the mixture were analyzed to compare the mixing effect of the variable-diameter swirl shear mixer with that of the static mixer. The ejector was optimized based on the injection method to form a coupled static micromixer; furthermore, the mixing effect, flow field movement, and pressure drop of the two were compared using CFD numerical simulation calculations. Compared with other static mixers, coupled static micromixers can reduce the median particle size of dispersed phase from 39 μm to 41 μm to 23 μm, which is equivalent to the mixing effect of the dynamic high-shear mixer at the speed of 2100 r/min to 2400 r/min. It has the characteristics of simple structure and uniform dispersion effect. Therefore, the proposed coupled static micromixer has broad application prospects.

2. Experiment and simulation

2.1. Experimental

2.1.1. Experimental device

A schematic of the experimental device is shown in Fig. 1; the device mainly comprises a compressor (Otus OTS-800; nominal volume flow, 60 L/min; rated exhaust pressure, 0.7 MPa; speed, 1380 r/min), pressure vessel storage tank (volume, 10 L), booster water pump (Ville MHI205N-1/10/E/3-380-50-2; maximum flow, 5 m^3/h; maximum head, 54 m), water tank, glass rotor flowmeter (maximum range, 800 L/h), metal tube float flowmeter (CELAI Lion Automation; range, 0–200 L/h), and sewage tank. In the experiment, diesel oil was injected into the pressure vessel storage tank. Compressed gas was injected into the pressure vessel storage tank to create a certain pressure in the storage tank. The oil phase flow could be adjusted by adjusting the pressure and valve opening, the water phase flow could be adjusted by the pressure of the booster pump, and finally, a sample was recorded at the outlet three-way valve.

Both the swirl shear mixer and ejector adopted a three-dimensional (3D) printing structure. The swirl shear mixer was designed using the traditional jet mixer, variable diameter structure as well as the positive and negative spiral structures of the static mixer. A shear jet with a variable diameter structure was used in the ejector structure. A three-way structure was used outside the ejector. The water phase entered through another inlet. Thus, the swirling shear mixer and ejector formed a coupled static micromixer.

The commercial static mixers used in the experiment were SV and SH static mixers (as shown in Fig. 6(b)) produced by Delta Mixers, the static mixers were selected by optimizing and screening under experimental conditions. The interior of the SV mixer comprises interlaced corrugated sheets: the angle between the corrugation and the pipe axis is 45°, and the angle between the adjacent corrugated sheets is 90°. The SH static mixer comprises only repeating mixing units and within each mixing unit, there are two spiral pieces inside; furthermore, there is a mixing chamber between adjacent mixing units. The high shear mixer was produced using a Shanghai Yiken high shear emulsifier.

2.1.2. Experimental system

In this study, a diesel/water heterogeneous system was used to evaluate the mixing effect. The surface tension of diesel was 26.8 mN m^{-1}, the viscosity was 3.87 mPa s at 25 °C, and the interfacial tension was determined by the United States Kono Industries (USA Kono Industry Co. Ltd; optical contact angle and interface tension meter) measured by the optical contact angle and interface tension meter. Viscosity was measured using a rotational viscometer. Tap water from the laboratory was used (Table 1).

2.2. Simulation

2.2.1. Physical model and meshing

Fig. 2(b) shows the 3D printing model. The model was created using SolidWorks (SW) according to the parameters of the experimental model. The physical model used in the simulation was identical in size to the structure of the swirl and coupled static micromixers in the laboratory. The fluid of the model was extracted using the combined function provided in SW. After the fluid extraction, the ICEM was used to divide the physical model. Because of the complexity of the spiral structure, a tetrahedral mesh was used. Finally, using grid independence verification, we determined that the number of grids in the static micromixer was 2089907. The coupled static micromixer possessed a part for a small injection; therefore, the local area was encrypted, and the final number of grids was 280090.

2.2.2. Setting of the boundary condition and calculation method

The mixture multiphase flow model and the standard k-ε turbulence model were used to analyze the flow field, velocity field, and pressure field of the static and compound static micromixers in a turbulent state. The standard k-ε model is widely used because it provides stable solutions and high accuracy.

The continuity equation of the mixture model is as follows:

$$\frac{\partial \rho_m}{\partial t} + \nabla \bullet (\rho_m \overline{\upsilon}_m) = 0 \quad (1)$$

$$\overline{\upsilon}_m = \frac{\sum_{k=1}^{n} \Phi_k \rho_k \overline{\upsilon}_k}{\rho_m} \quad (2)$$

$$\rho_m = \sum_{k=1}^{n} \Phi_k \rho_k \quad (3)$$

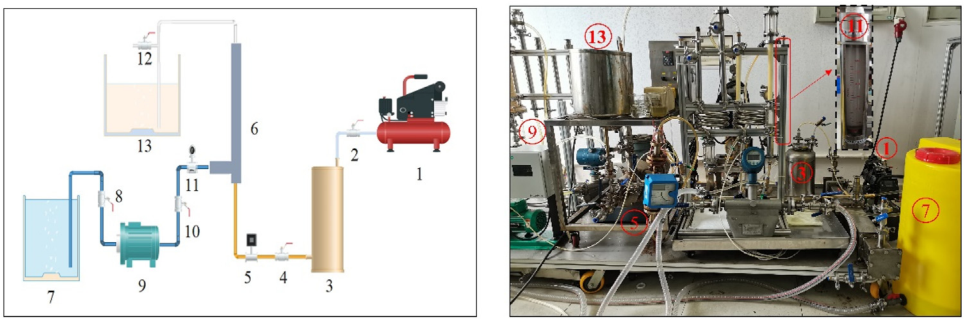

Fig. 1 – Schematic diagram of the experimental device.
1 air compressor; 2, 4, 8, 10 valves; 3 pressure vessel tank, 5 rotameter; 6 micromixer/composite micromixer; 7 water tank; 9 booster pump; 11 rotameter; 12 three-way valve; 13 sewage tank.

Table 1 – Physical parameters.

Phase	Parameter		
	Density ρ/(Kg/m^3)	Viscosity μ/(mPa s)	Surface tension σ/(mN/m)
Water	998	1.02	72.8
Diesel	842	3.87	26.8

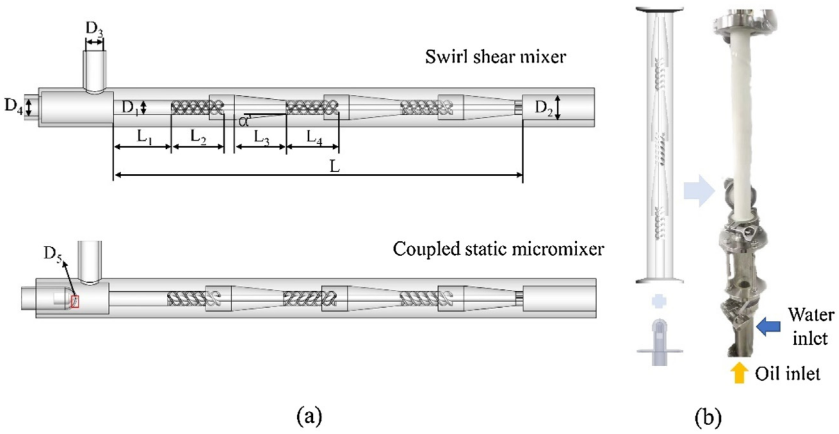

Fig. 2 – Simulation structure diagram: (a) model diagram, (b) 3D printing combination diagram.

where ρ_k is the density of the kth phase, $\overline{\upsilon_k}$is the average velocity of the kth phase, and ϕ_k is the volume fraction of the kth phase.

The momentum equation of the mixture model can be obtained by a summation of the respective momentum equations for all phases as:

$$\frac{\partial}{\partial t}(\rho_m \overline{\upsilon}_m) + \nabla \bullet (\rho_m \overline{\upsilon}_m \overline{\upsilon}_m) = -\nabla p + \nabla \bullet [\mu_m (\nabla \overline{\upsilon}_m \overline{\upsilon}_m + \nabla \overline{\upsilon}_m^T)] + \rho m g + F \tag{4}$$

where m is the number of phases, F is the volume force, g is the acceleration due to gravity, and μ_m is the mixed viscosity.

$$\mu_{\text{m}} = \sum_{k=1}^{n} \Phi_k \mu_k \tag{5}$$

The volume fraction equation of the phase P is:

$$\frac{\partial}{\partial t}(\phi_P \overline{\upsilon}_P) + \nabla \bullet (\phi_P \overline{\upsilon}_P \overline{\upsilon}_m) = -\nabla \bullet \left(\phi_P \overline{\upsilon}_P \overline{\upsilon}_{dr,p}\right) \tag{6}$$

In the simulation, a velocity inlet was inserted at the inlet of the fluid. The water phase was considered as the primary phase, diesel as the dispersed phase; there was no interphase slip velocity. The volume fraction of the second phase at the inlet of the water phase was 0, and the volume fraction of the second phase at the inlet of the oil phase was 1; this implies that the water and oil phase inlets comprised completely of water and diesel, respectively. The main phase flow rate was 1 m/s, the dispersed phase flow rate was 0.1 m/s, and the outlet was considered as the pressure outlet. Ansys FLUENT adopts a three-dimensional single-threaded calculation and uses a pressure-based solver and the SIMPLE algorithm to solve. The main parameters and momentum volume fraction equation in the solver adopted the QUICK second-order format, and the residual 10×10^{-5} was used as the monitor of the iterative calculation.

3. Results and discussion

3.1. *Design and simulation research on the coupled of ejector and swirl shear mixer*

3.1.1. *Concentration field analysis*

In the numerical simulation, the distribution of the concentration field directly reflects the mixing degree of the two phases. Fig. 3(a) and (b) shows contrast contours of the vol-

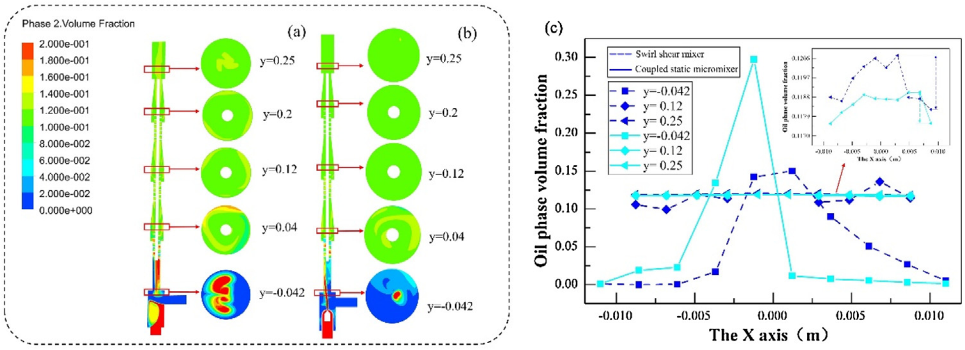

Fig. 3 – Concentration distribution contours of (a) swirl shear mixer and (b) coupled static micromixer; (c) concentration distribution at different positions at X-axis.

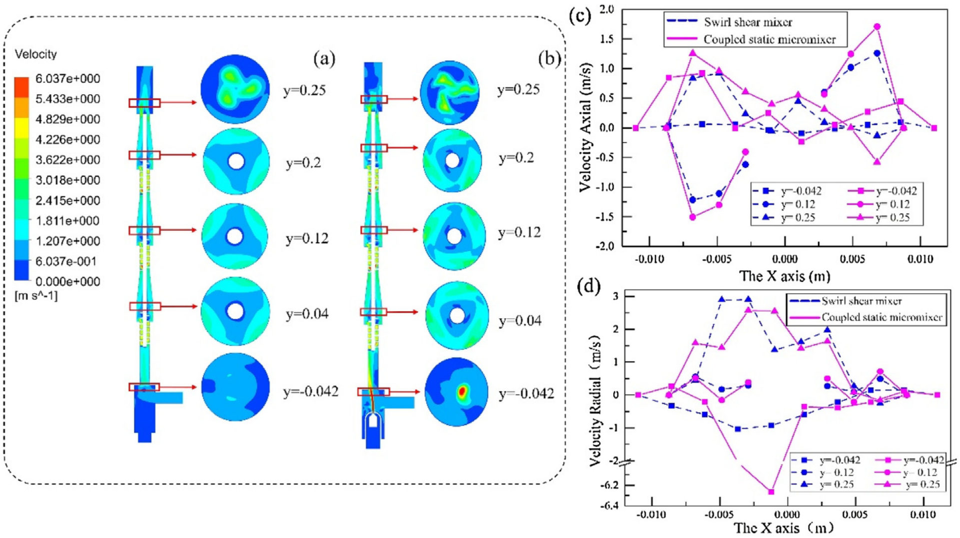

Fig. 4 – Velocity field contours of the (a) swirl shear mixer and (b) coupled static micromixer; (c) axial velocity and radial velocity at different positions at X-axis.

ume distribution of the oil phase at different cross-sections. After passing through the first swirl (y = 0.04 m), the phenomenon of uneven mixing of the two phases continued to manifest, wherein the highest concentration was 18%, and the distribution was not uniform (shown in Fig. 3(a)); As shown in Fig. 3(b), the highest concentration at the aforementioned position was 14%. After passing through the second swirl part (Y = 0.12 m), uneven parts continued to exist at the matching position in Fig. 3(a); meanwhile, the coupled static micromixer was mixed evenly; however, the final outlet part of the swirl shear mixer was remainedpartly uneven. Fig. 3(c) show the concentration distributions on the x-axis for different cross-sections. In Fig. 3(c), the oil phase volume distribution of the swirl shear mixer is still uneven at y = 0.012 m, with the smallest oil phase concentration distribution being 10% and the largest being 14%; the oil phase concentration difference of the dispersed phase is still greater than that of the coupled static micromixer at y = 0.25 m. This is because the coupled static micromixer has the effect of shearing and strengthening the dispersion of the oil phase due to its ejector, which greatly improves the oil and water mixing effect.

3.1.2. *Velocity field analysis*

As seen from Fig. 4(a) and (b), subsequent to the primary phase fluid and dispersed phase fluid entering the inlet, the velocity increases through the spiral section. As the diameter abruptly increases, the velocity decreases rapidly. The speed close to the center is lower and the speed close to the outer wall is high. In the tapered section, the fluid speed follows a trend wherein it gradually tends to be balanced. The entire cross-section flow field is composed of many vortices and the energy of the fluid is predominantly distributed in the vortices. Because of the velocity difference between the large and small vortices, a shear force is generated between the fluids, increasing the turbulent kinetic energy, which is conducive to mixing the two phases. In the swirling flow field, owing to the change in the velocity gradient, different centrifugal forces and shear forces are caused by the flow field. Therefore, the abrupt change in the turbulence intensity in the flow field and the central support column prevents the oil phase from migrating to the center owing to centrifugation, which is conducive to the collision and dispersion between the water and oil phase droplets and improves the process of mixing and dispersion.

Owing to the spiral blades, the fluid goes a swirling motion caused by the action of the centrifugal field after passing. As shown in Fig. 4(c), in the spiral section, the axial velocity is inverse symmetry. The oil droplets with each other exhibit turbulent collisions owing to the opposite velocity directions, thus, improving the mixing of oil and water. As shown in Fig. 4(d), the radial velocity variation of the coupled static micromixer at the mixing part (y = 0.12 m) is greater than that of the swirl shear mixer, resulting in a larger radial shear force of the fluid and a stronger shearing and crushing effect on the oil phase. At y = −0.042 m, the radial velocity of the coupled

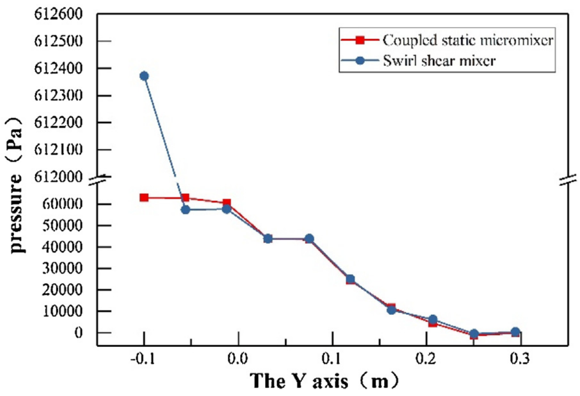

Fig. 5 – Y-axis pressure at different positions of swirl shear mixer and coupled static micromixer.

static micromixer is much higher than that of the swirl shear mixer. The reverse direction of the velocity is due to a shrinking change at the cross section, which leads to reflux of the fluid. Because there is a water inlet on the right side, the jet portion of the fluid tends to drift to the left. The velocity near both walls is close to 0 m/s, which is due to the viscosity of the fluid, resulting in a decrease in velocity near the walls. In the coupled static micromixer, the strengthening of velocity field will produce stronger turbulence intensity, and the collision and dispersion of droplets in the flow field will enhance the dispersion effect, which is conducive to the strengthening of mixing effect. The magnitude of the turbulent kinetic energy is an important indicator of the degree of mixing, the greater the turbulent kinetic energy, the greater the turbulence intensity of the two-phase mixing, which will strengthen the collision and dispersion of droplets.

3.1.3. Pressure drop analysis

Pressure drop is an important indicator for static mixers; it directly determines the energy consumption and has a significant influence on the operation performance and mixing effect. Owing to the different internal mixing elements, the pressure drops of the two are different. Additionally, for the static micromixer, pressure loss is an important parameter for evaluation. In practical applications, if the mixing effect of the equipment is acceptable, but the head loss pressure is large, the selection and practical application of a static micromixer are not used. As seen in the static pressure distribution chart (Fig. 5), the pressure at the front end of the swirl shear mixer is the largest, and the whole shows an abrupt drop. This is because when the fluid passes through the first spiral and the tapered sections, owing to the reduction in the cross-sectional area, the fluid accelerates the movement and converts part of the static pressure into dynamic pressure. Subsequent to this repetition, the pressure drops in steps at each instant that a spiral section and a tapered section pass. The coupled static micromixer has a large change in the cross-sectional area at the front ejector; therefore, the pressure changes significantly and the maximum pressure is 6.124×10^5 Pa.

3.2. Effect comparison between swirl shear mixer and traditional static mixer

After employed an optimized selection, SH and SV static mixers were selected and their mixing effect was compared to that of swirl shear mixers, under conditions where the main phase water flow was 500 L/h, and the dispersed phase flow was 30 L/h. As shown in Fig. 6(a), the median particle size produced by the swirl shear, SV static and SH static mixers was 29 μm, 39 μm, and 41 μm, respectively. Based on the lower accumulation curve of the swirl shear mixer, increased concentration of the particle size distribution was observed, which indicates that the dispersion effect of the oil droplets particle size of the swirl shear mixer was far better than that of the SV and SH static mixers. The Fig. 6(b) shows the turbidity sedimentation distributions of the three mixers. Immediately after sampling, the sampling result of the swirl shear mixer was larger than the measurement range of the turbidity meter; therefore, there was no numerical display. The turbidity values of the SH and SV static mixers were smaller than those of the swirl shear mixer, which indicates that the mixing effect of the swirl shear mixer is better than the other two. As the settling time increased, the slope of the turbidity sedimentation curve gradually approached matching level. The clarification rate of the entire turbidity sedimentation curve was not a fixed value; this observation agrees with Stokes sedimentation law. The particle size and sedimentation velocity exhibited a positive quadratic relationship. The turbidity sedimentation curve can be changed to another degree of reaction mixing degree and particle size distribution.

Although the results of the mixing effects of the swirl shear mixer were better than that of the SH and SV static mixers, there is room for improvement regarding the injection method of the mixer. The conventional mixer injection method is the primary phase and the dispersed phase into the mixing element. At a certain distance inside the mixer, the two phases are pre-mixed and gradually mixed in-depth, which requires the mixer to be lengthy. Therefore, in this study, an ejector structure was designed, which could be premixed in the ejector part and improve the degree of oil-water two-phase mixing and significantly shorten the length of the mixer.

3.3. The design and dispersion effect of the ejector

The choice of the ejector has a significant influence on the mixing effect of the coupled static micromixer. If the injection hole diameter is too large, the mixing effect will be poor. If the injection hole diameter is too small, the pressure drop in the ejector will be too large. Additionally, for the mixer, pressure loss is an important part of the evaluation of the mixer. Therefore, an ejector with an injection hole diameter of 0.8 mm was selected after experimental optimization. Fig. 7(d) shows the pressure from 0.2 MPa to 0.5 MPa. Under a pressure of 0.2 MPa, the micro-dispersed phase was granular; however, with an increase in pressure, the dispersion effect of the dispersed phase improved, indicating a mist state.

After optimization, the ejector aperture was 0.8 mm to form a coupled static micromixer; the swirl shear mixer and coupling type were compared at a water phase flow rate of 500 L/h and an oil phase flow rate of 30 L/h. As shown in the oil droplets particle size distribution in Fig. 8(a), the median particle size of the coupled static micromixer was found to be 29 μm, and the maximum particle size was 100 μm, while the median particle size of the coupled static micromixer was 24 μm, and the maximum particle size was approximately 45 μm, which is far better than the dispersion effect of the swirl shear mixer. The Fig. 8(b) shows the turbidity sedimentation curve of the two over time. The turbidity decline rate of the two was similar in the beginning 10 min. This is because both of them have large the oil droplets that float upward quickly, resulting in rapid clarification of the sample. With time, the clarification

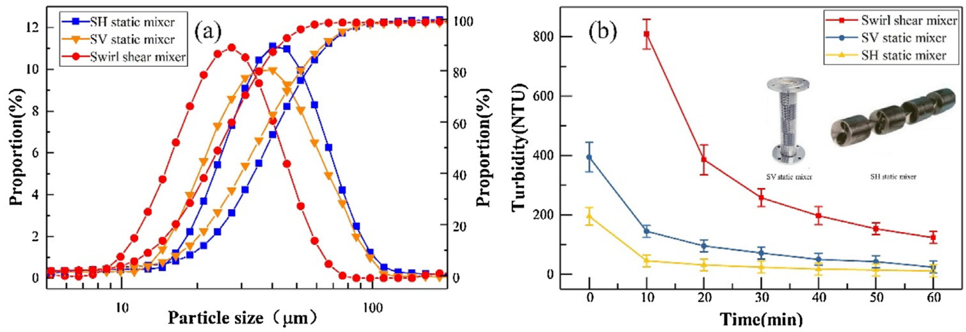

Fig. 6 – The (a) particle size distribution and (b) turbidity diagram of the swirl shear mixer and the coupled static micromixer.

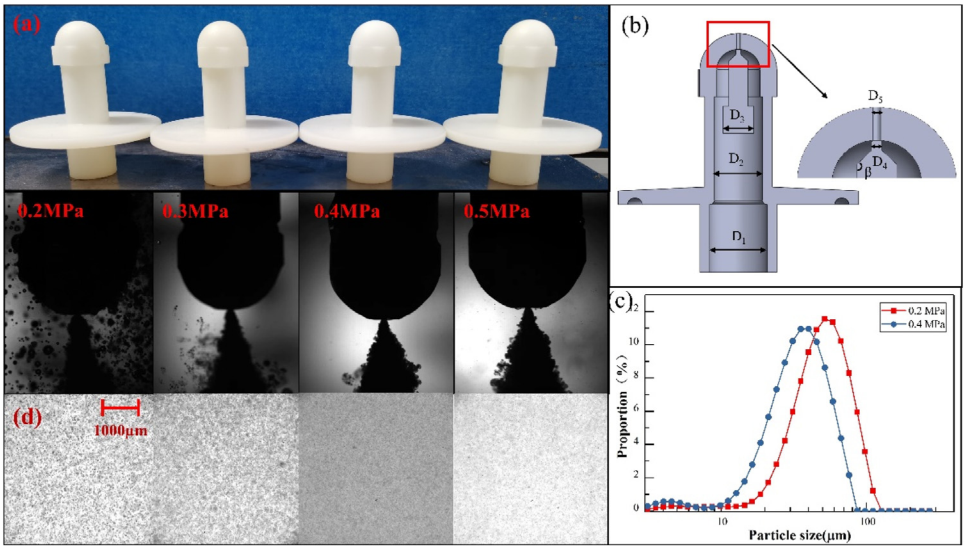

Fig. 7 – Injection diagram of the ejector at different pressures.

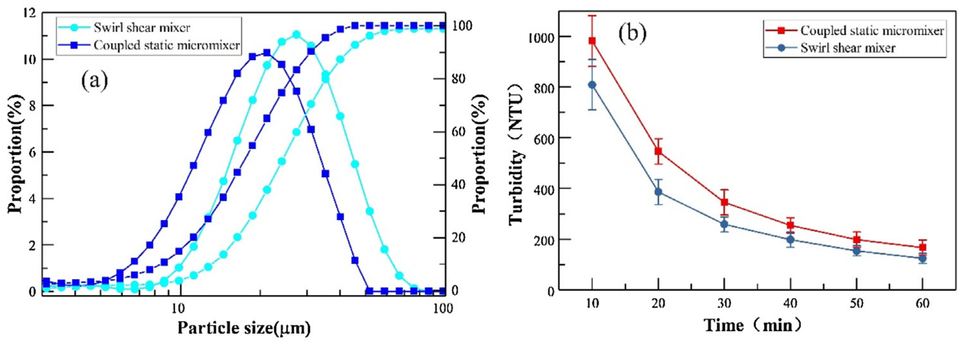

Fig. 8 – The (a) particle size distribution and (b) turbidity diagram of the swirl shear mixer and the coupled static micromixer.

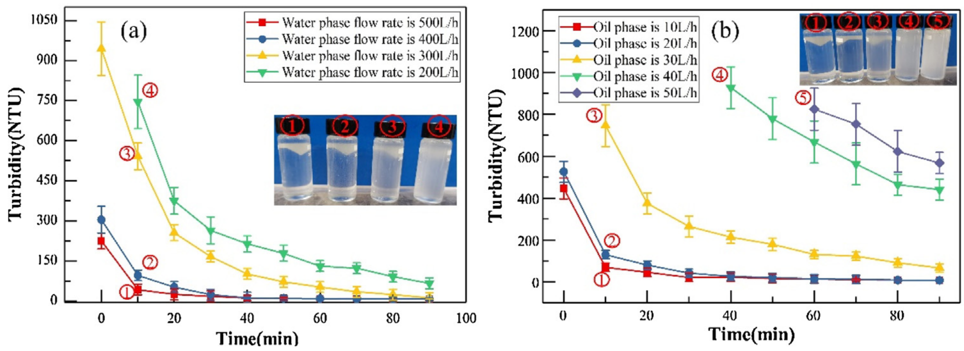

Fig. 9 – Turbidity diagram at (a) different flow rates and (b) different phases rate of the coupled static micromixer.

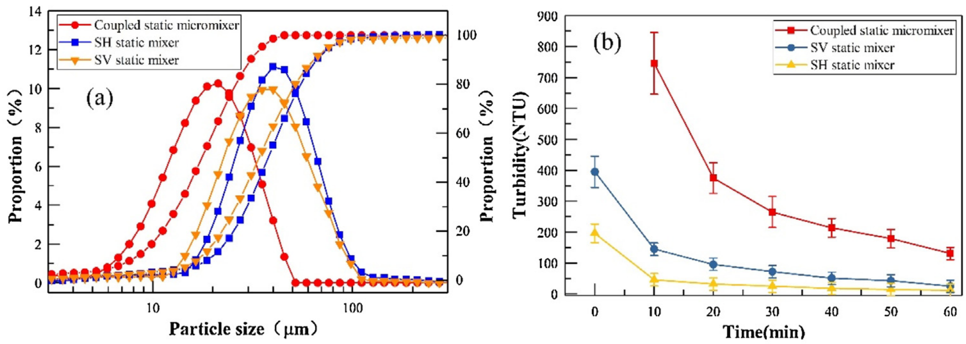

Fig. 10 – The (a) particle size distribution and (b) turbidity diagram of coupled static micromixer, SH static mixer, SV static mixer.

rate of the coupled static micromixer tended to be flat, and the turbidity remained 200 NTU at 60 min.

3.4. *The mixing effect of the coupled static micromixer*

The flow rate and two-phase ratio of the coupled static micromixer are also the main factors that affect the mixing effect. A small flow rate leads to a weak internal flow field strength and insufficient mixing strength. The Fig. 9(a) shows the turbidity sedimentation curve when the oil/water ratio was 6%, and the water phase flow rates were 500, 400, 300, 200, and 100 L/h. With an increase in the flow rate, the turbidity continuously increased. There was a clear difference in the turbidity between the water phase flow rates of 400 and 300 L/h. When the structural parameters of the coupled static micromixer were fixed, the flow rate had a significant influence on the mixing situation. The larger the flow, the more violent the flow field movement inside the coupled static micromixer, and the better the mixing effect. At a water flow rate of 500 L/h, the slope of the sedimentation curve was the slowest. According to Stokes' law, the smaller the particle size, the slower the oil droplets rising speed and the slower the clarification rate. The Fig. 9(b) shows the turbidity sedimentation curves at 2%, 4%, 6%, 8%, and 10% when the water phase flow was 500 L/h and the oil/water ratios were 2%, 4%, 6%, 8%, and 10%. When the water/water ratio was 2% and 4%, the turbidity decreased significantly. After 10 min, it fell below 200 NTU, and it was close to a clear state after 80 min. When the oil/water ratio was 6%, the turbidity started to level off at 50 min, and the turbidity was still around 100 NTU at 80 min.

3.5. *Comparison of the effect of coupled static micromixer, traditional static mixer, and the dynamic high-shear mixer*

To compare and analyze the turbidity, when oil/water was at 6% and the water phase flow was 500 L/h, the coupling static micro-mixer, and SH and SV static mixers were selected to compare the mixing effect. As shown in Fig. 10(a), the median particle size produced by the coupled static micromixer was 23 μm, the median particle size of the SV static mixer was 39 μm, and the median particle size of the SH was 41 μm. Based on the lower accumulation curve of the coupled static micromixer, the particle size distribution is more concentrated, which shows that the dispersion effect of the oil phase particle size in the coupled static micromixer is far better than that of the SV and SH static mixers. The Fig. 10(b) shows the turbidity sedimentation distributions of the three mixers. The turbidity value directly shows that the mixing effect of the coupled static micromixer is better than that of the two. As the sedimentation time increased, the slope of the turbidity sedimentation curve gradually tended to level.

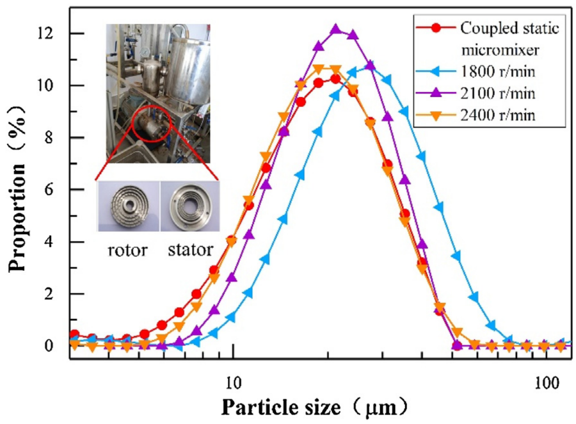

Fig. 11 – The particle size distribution between the dynamic high-shear mixer and coupled static micromixer.

Under identical parameters and conditions, the mixing effect of the dynamic high-shear mixer at different speeds and the coupled static micromixer were compared. The dynamic high-shear mixer adjusts the speed according to the frequency, after the internal rotor and stator mesh, a sample was taken at the outlet to measure the particle size and turbidity. As shown in Fig. 11, the median particle size of the static micromixer ranges from 2100 to 2400 r/min for the dynamic high shear mixer.

4. Conclusion

This work has optimized the injection method of the dispersed phase. The dispersed phase is injected in a spray mode, which significantly improves the mixing effect of the mixer. The study compared the mixing effects before and after optimization and the mixing effects of conventional static mixers and dynamic high-shear mixers. The following conclusions are drawn based on the results of the experiments that we conducted.

(1) The effects of the swirl shear mixer are compared with the traditional SH and SV static mixers. Under the conditions of a water flow of 500 L/H and oil flow of 30 L/H, the median particle size of oil droplet in the swirl shear mixer was 29 μm, and the median size of the oil droplets in the SH and SV static mixers are 41 mm and 39 mm, respectively.

(2) The injection method of the dispersed phase in the swirl shear mixer is improved. The CFD simulation and experiment are compared. The median particle size (23 μm) and turbidity settling time of the coupled static micromixer are better than those of the single swirl shear mixer.
(3) The turbidity settling curves were compared when the water phase flow was 500 L/h, the oil-water ratio was 2%–10%, and the oil-water ratio was 6%, the water phase flow was 200 L/h–500 L/h. The larger the oil-water ratio and water phase flow, the larger the turbidity.
(4) The results show that the mixing effect of the coupled static micromixer is much better than that of the static SH and SV mixers at a water flow rate of 500 L/h and an oil-water ratio of 6%. The particle size distribution of the coupled static micromixer was between 2100 and 2400 rpm for the dynamic high-shear mixer. The accuracy of the results was verified by the turbidity sedimentation curve.

Declaration of interests statement

There is no conflict of interest in this paper.

Declaration of Competing Interest

The authors report no declarations of interest.

Acknowledgments

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